



K A G E

The Problem

The Problem

A Million-Euro Blind Spot

Disruption from Above

Every year, solar storms interfere with radio signals and GPS navigation, costing polar airlines **hundreds of millions of euros** in extra fuel and lost revenue.

- ✓ **€400,000 cost** per single avoidable diversion.
- ✓ **30-40 days** of significant HF disruption annually.
- ✓ Legacy policy manuals force dispatchers to "guess."

Modern Cockpit Display

The Value Pro

A Milion-Euro Blind Spot



The Value

A Milion-Euro Blind Spot

```
FNXX01 YMMC 020100
SWX ADVISORY
DTG:                20190202/0100Z
SWXC:               ACFJ
ADVISORY NR:        2019/10
SWX EFFECT:         HF COM MOD
OBS SWX:            02/0100Z DAYLIGHT SIDE
FCST SWX + 6 HR:    02/0700Z DAYLIGHT SIDE
FCST SWX + 12 HR:   02/1300Z DAYLIGHT SIDE
FCST SWX + 18 HR:   02/1900Z NO SWX EXP
FCST SWX + 24 HR:   03/0100Z NO SWX EXP
RMK:                LOW END OF BAND HF COM DEGRADED
                   ON SUNLIT ROUTES. NEXT 12 HOURS
                   MOST POSSIBLE, DECLINING THEREAFTER.
NXT ADVISORY:       20190202/0700Z=
```

" The reason for your flight delay is often up there. Not the weather. Not the engine. The sun.

— ASTROROUTE MISSION STATEMENT

Actionable Intelligence

From Data to Decisions

We ingest live satellite feeds from NOAA and ESA, translating complex space weather bulletins into actionable rerouting options in **under 60 seconds**.

- ✓ **3 Ranked Routes:** Optimised for cost and safety.
 - ✓ **Risk Scoring:** 200km route segment granularity.
 - ✓ **Decision Support:** Dispatcher retains final authority.
-



365-Day Value Proposition



Polar Safety

Protecting passengers and crew from radiation exposure and navigation loss during major events.



ESG Compliance

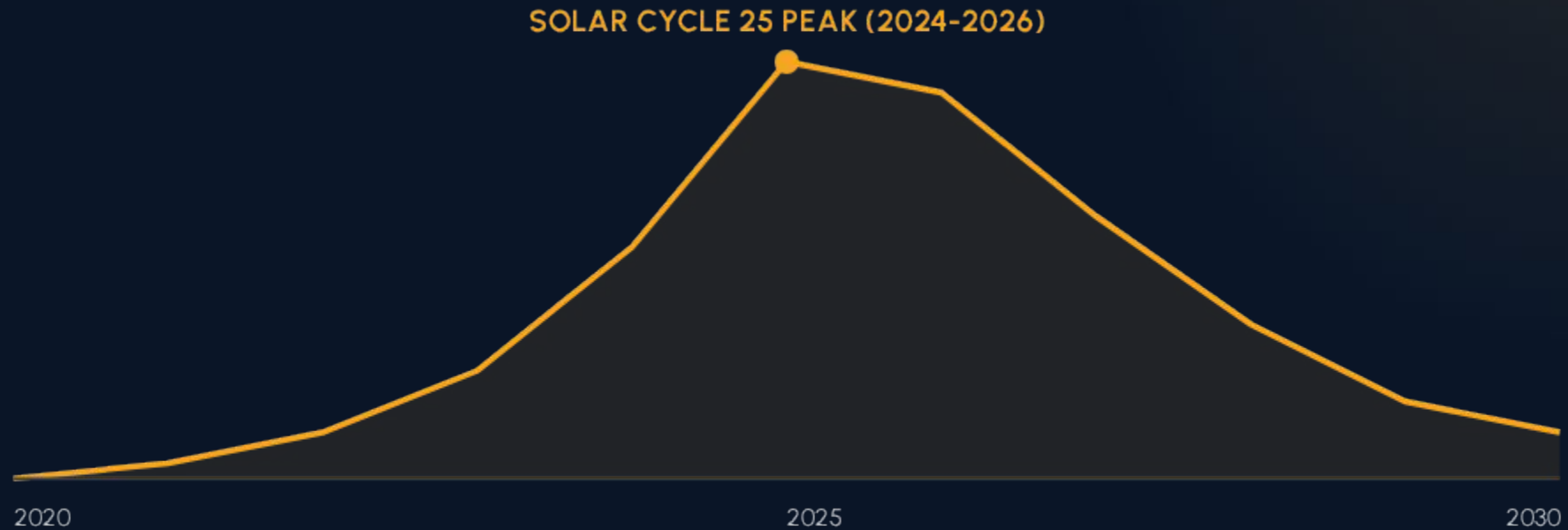
Every tonne of fuel saved equals **3.15 tonnes of CO2** avoided.
Automated sustainability reporting.



Daily Savings

Optimising routes every single day, cutting fuel costs regardless of solar activity levels.

Solar Maximum: Peak Opportunity



We are currently at the peak of solar activity. The window for airlines to adopt risk mitigation is **now**.

Validating the Industry

COMPETITION & DIFFERENTIATION

TRADITIONAL ALERTS

ICAO / PECASUS Bulletins

- Generic, text-based
- Manual interpretation
- No fuel delta analysis
- Safety only focus

KAGE OPTIMIZER

Decision Support Engine

- Asset-specific recommendations
- Automated ML integration
- Fuel vs. Risk optimization
- Saves money while ensuring safety

Go-to-Market Strategy

1

Phase 1: LAND

3 Paid Pilots with Finnair, Lufthansa, and SAS. Secure ESA BASS funding.

2

Phase 2: PROVE

Publish ROI case studies. Convert pilots to annual contracts. Launch API track.

3

Phase 3: EXPAND

12+ Direct contracts. Global vendor API integration. Full ESG reporting suite.

The Multi-Disciplinary Team

0 LEGACY BIAS.
100% DEPTH.



Gonzalo

Astrophysics & Aerospace Engineering



Kornelia

Biomedicine & Radiation Safety



Eric

Mechanical Engineering & Finance



Álvaro

Astrophysics & Aerospace Engineering

Scalable Subscription Model

€270M

Total Addressable Market

Subscription Tiers

- ✓ **Starter:** Alert Dashboard (€15k-25k/yr)
- ✓ **Core:** Route Optimization (€60k-150k/yr)
- ✓ **Platform:** API Vendor Partnerships (Rev Share)

THE SKY ABOVE 60°N IS A BILLION-EURO BLIND SPOT.

WE BUILT THE MAP.

The Ask: €600,000

18-month runway to achieve €1.2M ARR. Join us at the peak of Solar Cycle 25.

astroroute.space

StarTech Summer School 2026

Group F | Investor Pitch

Image Sources



https://upload.wikimedia.org/wikipedia/commons/3/3d/Great_Circle-Rhumb_Line_Map_Animation.gif

Source: en.wikipedia.org



<https://www.flyingmag.com/wp-content/uploads/sites/2/2021/08/httpswww.flyingmag.comsitesflyingmag.comfilesimages201703head-up-display-1.jpg>

Source: www.flyingmag.com



<https://robertwilkos.com/wp-content/uploads/2025/11/Inside-the-247-Operations-Center-How-Elite-Aviation-Companies-Deliver-Flawless-Service.jpg>

Source: robertwilkos.com



<https://www.popsci.com/wp-content/uploads/2025/08/solar-flare-DKIST.png?quality=85&w=2000>

Source: www.popsci.com



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Thumbnail

for

[www.lemon8-](https://www.lemon8-app.com)

[app.com](https://www.lemon8-app.com)

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